Gen Li

li.gen@ed.ac.uk | Homepage | Github | Google Scholar

EDUCATION

University of Edinburgh *2021.09 - 2025.07 (Expected)*
Ph.D. Robotics and Autonomous Systems - Edinburgh, UK

Sungkyunkwan University *2018.09 - 2020.08*
M.S. Electrical and Computer Engineering - Suwon, South Korea

Xidian University *2014.08 - 2018.06*
B.S. Electronic and Information Engineering - Xi'an, China

RESEARCH FIELDS

- **Embodied AI** - Affordance Learning, Robotic Manipulation
- **Computer Vision** - Multi-Modal Learning, Vision-Language Models
- **Machine Learning** - Few-Shot Learning, Learning under Limited Supervision

RESEARCH EXPERIENCE

Visual Affordance and Object Functionality Understanding *2021.09 - Present*
PhD's Research, Supervisor: Dr. Laura Sevilla-Lara

Prototypical Learning for Few-Shot Semantic Segmentation *2020.10 - 2021.08*
Visiting Researcher at the UoE, Supervisor: Dr. Laura Sevilla-Lara

Neural Networks for Real-time Semantic Segmentation *2019.06 - 2020.07*
Master's Research, Supervisor: Prof. Joongkyu Kim

Semantic Image Segmentation based on Deep Learning *2017.11 - 2018.06*
Undergraduate Thesis, Supervisor: Prof. Cheolkon Jung

SELECTED PUBLICATIONS (total citations: 900+)

- Learning Precise Affordances from Egocentric Videos for Robotic Manipulation [project]
  Gen Li, Nikolaos Tsagkas, …, Sethu Vijayakumar, Kun Shao, Laura Sevilla-Lara
  In Submission
- One-Shot Open Affordance Learning with Foundation Models [project]
  Gen Li, Deqing Sun, Laura Sevilla-Lara, Varun Jampani
  IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024
- LOCATE: Localize and Transfer Object Parts for Weakly Supervised Affordance Grounding
  Gen Li, Varun Jampani, Deqing Sun, Laura Sevilla-Lara [project]
  IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- Adaptive Prototype Learning and Allocation for Few-Shot Segmentation [project]
  Gen Li, Varun Jampani, Laura Sevilla-Lara, Deqing Sun, Jonghyun Kim, Joongkyu Kim
  IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021
  [390+ Citations]
- DABNet: Depth-wise Asymmetric Bottleneck for Real-time Semantic Segmentation [pdf] [code]
  Gen Li, Joongkyu Kim
  British Machine Vision Conference (BMVC), 2019
  [300+ Citations]
- Depth-wise Asymmetric Bottleneck with Point-wise Attention Decoder for Real-time Semantic Segmentation in Urban Scenes [pdf]
  Gen Li, Shenlu Jiang, Inyong Yun, Jonghyun Kim, Joongkyu Kim
  IEEE Access (SCI, IF: 4.098)

- Weakly Supervised Temporal Attention 3D Network for Human Action Recognition [pdf]
  Jonghyun Kim, **Gen Li**, Inyong Yun, Cheolkon Jung, Joongkyu Kim
  Pattern Recognition (SCI, IF: 7.196)
- Edge and Identity Preserving Network for Face Super-Resolution [pdf] [code]
  Jonghyun Kim, **Gen Li**, Inyong Yun, Cheolkon Jung, Joongkyu Kim
  Neurocomputing (SCI, IF: 4.438)
- Referenceless User Controllable Semantic Image Synthesis [pdf]
  Jonghyun Kim, **Gen Li**, Joongkyu Kim
  International Joint Conference on Neural Networks (IJCNN), 2023
- SuperStyleNet: Deep Image Synthesis with Superpixel Based Style Encoder [pdf] [code]
  Jonghyun Kim, **Gen Li**, Cheolkon Jung, Joongkyu Kim
  British Machine Vision Conference (BMVC), 2021
- An Optimized Deep Neural Network Detecting Small and Narrow Rectangular Objects in Google Earth Images [pdf]
  Shenlu Jiang, Wei Yao, Man-sing Wong, **Gen Li**, Zhonghua Hong, Tae-yong Kuc, Xiaohua Tong
  IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (SCI, IF:3.827)
- Perspective-Aware Density Regression for Crowd Counting [pdf]
  Yutong Wang*, **Gen Li***, Qi Zhang, Joongkyu Kim, Huifang Li
  IEEE International Conference on Image Processing (ICIP), 2021
- Progressive Face Super-Resolution with Non-Parametric Facial Prior Enhancement [pdf]
  Jonghyun Kim, **Gen Li**, Cheolkon Jung, Joongkyu Kim
  IEEE International Conference on Image Processing (ICIP), 2021

WORK EXPERIENCE

Huawei, Noah's Ark Lab, London *2023.07 - 2024.02*
Research Intern | Affordance-Oriented Robotic Manipulation

ACADEMIC SERVICE

Reviewer

- Conference – AAAI2024, ECCV 2024, CVPR2024
- Journal - IEEE Transactions on Image Processing, Neurocomputing

Teaching

- [2023 - 2024] Introductory Applied Machine Learning at UoE - Demonstrator and Tutor
- [2022 - 2024] Applied Machine Learning at UoE - Teaching Assistant
- [2018 - 2020] Digital Signal Processing at SKKU - Teaching Assistant

AWARDS AND GRANTS

- Ph.D. - EPSRC funded CDT-RAS Scholarship, 100% tuition waiver and an annual stipend
- M.S. - STEM Scholarship, 100% tuition waiver, Samsung Funding
- State Scholarship for Study Abroad, China Scholarship Council of P.R. China

PROFESSIONAL SKILLS

- Programming Language: Python, C/C++, LaTeX, Matlab, HTML/CSS, etc.
- Frameworks and Tools: Pytorch/Tensorflow/Keras, Git, Docker, Linux Bash, etc.